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06 — Kidney Filtration: Computational Physiology of GFR

BMED365 – Computational Medicine (Lab 5)

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Updated: 2026-02-20 (Cursor 2.5.20 with Claude Opus 4.6) - saved notebook to .pdf

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1. Motivation & Learning Objectives

Why study kidney filtration?

The **kidneys** are among the most vital organs in the human body. Despite weighing only about 150 g each (roughly 0.4% of body weight), they receive approximately **20--25% of the cardiac output** --- about 1.2 liters of blood per minute. Every day, the kidneys filter roughly **180 liters** of plasma through their glomeruli, producing about 1.5 liters of urine. This extraordinary filtration machinery maintains the body's internal environment by:

- **Removing metabolic waste** --- urea, creatinine, uric acid, drug metabolites
- **Regulating fluid balance** --- adjusting water reabsorption to maintain blood volume and osmolarity
- **Maintaining electrolyte homeostasis** --- precise control of Na^+, K^+, Ca^{2+}, phosphate, and H^+/HCO_3^-
- **Controlling blood pressure** --- through the renin-angiotensin-aldosterone system (RAAS) and fluid volume regulation
- **Endocrine functions** --- producing erythropoietin (red blood cell production), activating vitamin D, and synthesizing prostaglandins

The **glomerular filtration rate (GFR)** is the single most important measure of kidney function. A decline in GFR signals kidney disease, which affects **~10--15% of the global adult population** (over 800 million people). Understanding how GFR is measured, modeled, and estimated is therefore of immense clinical importance.

What you will learn

In this notebook you will:

1. **Understand** the anatomy of the kidney --- from the whole organ to the microscopic nephron --- and how structure enables function
2. **Appreciate** the historical journey --- from Carl Ludwig's filtration hypothesis (1842) to modern computational GFR estimation
3. **Master** the mathematical framework --- mass balance equations, the Starling equation, clearance concepts, and GFR estimation formulas
4. **Implement** a computational kidney filtration model in Python
5. **Visualize** and interpret filtration dynamics, mass balance, and the effects of pathological changes
6. **Explore** clinical applications --- CKD staging, drug dosing, dialysis, and non-invasive GFR estimation techniques (eGFR, MRI, nuclear medicine)

2. Historical Context: From Filtration Hypothesis to Computational GFR

The quest to understand urine formation

How does the kidney produce urine? This question puzzled physiologists for centuries. The answer emerged gradually through a combination of anatomical observation, physiological experimentation, and mathematical modeling.

Year	Milestone	Scientists
1666	First detailed description of kidney microstructure	Marcello **Malpighi**
1842	**Filtration hypothesis**: proposed that urine formation begins with mechanical filtration of blood through glomerular capillaries	Carl **Ludwig**
1844	Described Bowman's capsule; proposed secretion theory	William **Bowman**
1874	Demonstrated that urine composition depends on blood flow	Rudolf **Heidenhain**
1917	Concept of "clearance" as a measure of kidney function	Arthur **Cushny**
1926	First practical clearance measurements using creatinine	Poul **Rehberg**
1935	Established **inulin clearance** as the gold standard for GFR measurement	Homer **Smith**
1936	PAH (para-aminohippuric acid) clearance for renal plasma flow (RPF)	Homer **Smith**
1950s	Micropuncture studies confirming tubular function	A.N. **Richards** and others
1969	Cockcroft-Gault equation for estimated creatinine clearance	**Cockcroft** & **Gault**
1976	^{99m}Tc-DTPA and ^{51}Cr-EDTA for nuclear medicine GFR	Various groups
1999	**MDRD equation** --- estimated GFR from serum creatinine	Andrew **Levey** et al.
2009	**CKD-EPI equation** --- improved eGFR, now the clinical standard	Andrew **Levey** et al.

Year	Milestone	Scientists
2021	CKD-EPI equation updated to remove race coefficient	**Inker** et al.

Carl Ludwig and the filtration hypothesis

In 1842, the German physiologist **Carl Ludwig** (1816--1895) proposed a revolutionary idea: that urine formation begins as a purely physical process --- the mechanical **ultrafiltration** of blood plasma through the walls of the glomerular capillaries, driven by hydrostatic pressure. This was radical because it replaced vitalistic explanations with a quantitative, physics-based mechanism.

Ludwig's hypothesis was opposed by William Bowman, who favored a secretory theory. The debate persisted for decades until the micropuncture experiments of A.N. Richards in the 1920s--1940s directly confirmed that the fluid in Bowman's capsule is indeed an ultrafiltrate of plasma.

Homer Smith: the father of renal physiology

Homer W. Smith (1895--1962) at New York University established the modern quantitative framework for kidney function. His key contributions include:

- Demonstrating that **inulin** (a fructose polymer from chicory root) is freely filtered at the glomerulus, not reabsorbed, and not secreted --- making its clearance equal to GFR
- Developing **PAH clearance** to measure effective renal plasma flow (ERPF)
- Writing the definitive textbook *The Kidney: Structure and Function in Health and Disease* (1951)
- Establishing the conceptual framework of clearance that is still used in nephrology today

> *"The kidney is the most important organ in the body. The brain merely serves to carry the kidney around."* --- Homer W. Smith (half-jokingly)

From invasive to non-invasive GFR measurement

Historically, measuring GFR required invasive procedures: collecting timed urine samples over 24 hours while measuring plasma concentrations of a filtration marker (inulin or creatinine). These measurements are accurate but cumbersome, time-consuming, and impractical for routine clinical use.

The development of **estimated GFR (eGFR)** equations --- Cockcroft-Gault (1976), MDRD (1999), and CKD-EPI (2009, updated 2021) --- revolutionized nephrology by enabling GFR estimation from a **single blood test** (serum creatinine or cystatin C), combined with demographic variables (age, sex). Today, eGFR is automatically reported with every serum creatinine measurement in clinical laboratories worldwide.

Kidney anatomy cross-section

Figure 1. Cross-section of the human kidney showing the cortex, medulla (pyramids), renal pelvis, and ureter. Blood enters via the renal artery and exits via the renal vein. The functional unit — the **nephron** — spans cortex and medulla.
Source: Wikimedia Commons (public domain, by Piotr Michał Jaworski — PioM).

3. The Kidney: Biological Background

Gross anatomy

Each kidney is a bean-shaped organ (~12 cm long) located retroperitoneally on either side of the vertebral column. The key anatomical regions are:

- **Renal cortex** --- the outer region containing the glomeruli (where filtration occurs) and the proximal/distal convoluted tubules
- **Renal medulla** --- the inner region organized into cone-shaped **renal pyramids**, containing the loops of Henle and collecting ducts
- **Renal pelvis** --- a funnel-shaped cavity that collects urine from the collecting ducts and channels it to the ureter
- **Renal hilum** --- the entry/exit point for the renal artery, renal vein, and ureter

The nephron: functional unit of the kidney

Each kidney contains approximately **1 million nephrons**. Each nephron consists of:

Segment	Location	Primary function
Glomerulus + **Bowman's capsule**	Cortex	Ultrafiltration of plasma
Proximal convoluted tubule (PCT)	Cortex	Reabsorbs ~65% of filtered water, Na^+, glucose, amino acids
Loop of Henle (descending/ascending)	Medulla	Creates osmotic gradient; concentrating mechanism
Distal convoluted tubule (DCT)	Cortex	Fine-tuning of Na^+, K^+, Ca^{2+}; regulated by aldosterone
Collecting duct	Medulla to pelvis	Final water reabsorption (regulated by ADH/vasopressin)

The glomerular filtration barrier

The filtration barrier in the glomerulus is a three-layered structure:

1. **Fenestrated endothelium** --- capillary wall with small pores (~70 nm) that allow passage of water and small solutes but block blood cells
2. **Glomerular basement membrane (GBM)** --- a dense extracellular matrix that filters by size and charge (negatively charged, repelling albumin)
3. **Podocytes** (visceral epithelial cells) --- specialized cells with interdigitating foot processes forming **filtration slits** (~40 nm)

This barrier allows free passage of water, electrolytes, and small molecules (glucose, urea, creatinine) while retaining large proteins (albumin, MW ~66 kDa) and blood cells in the circulation.

The renal vasculature: a portal system

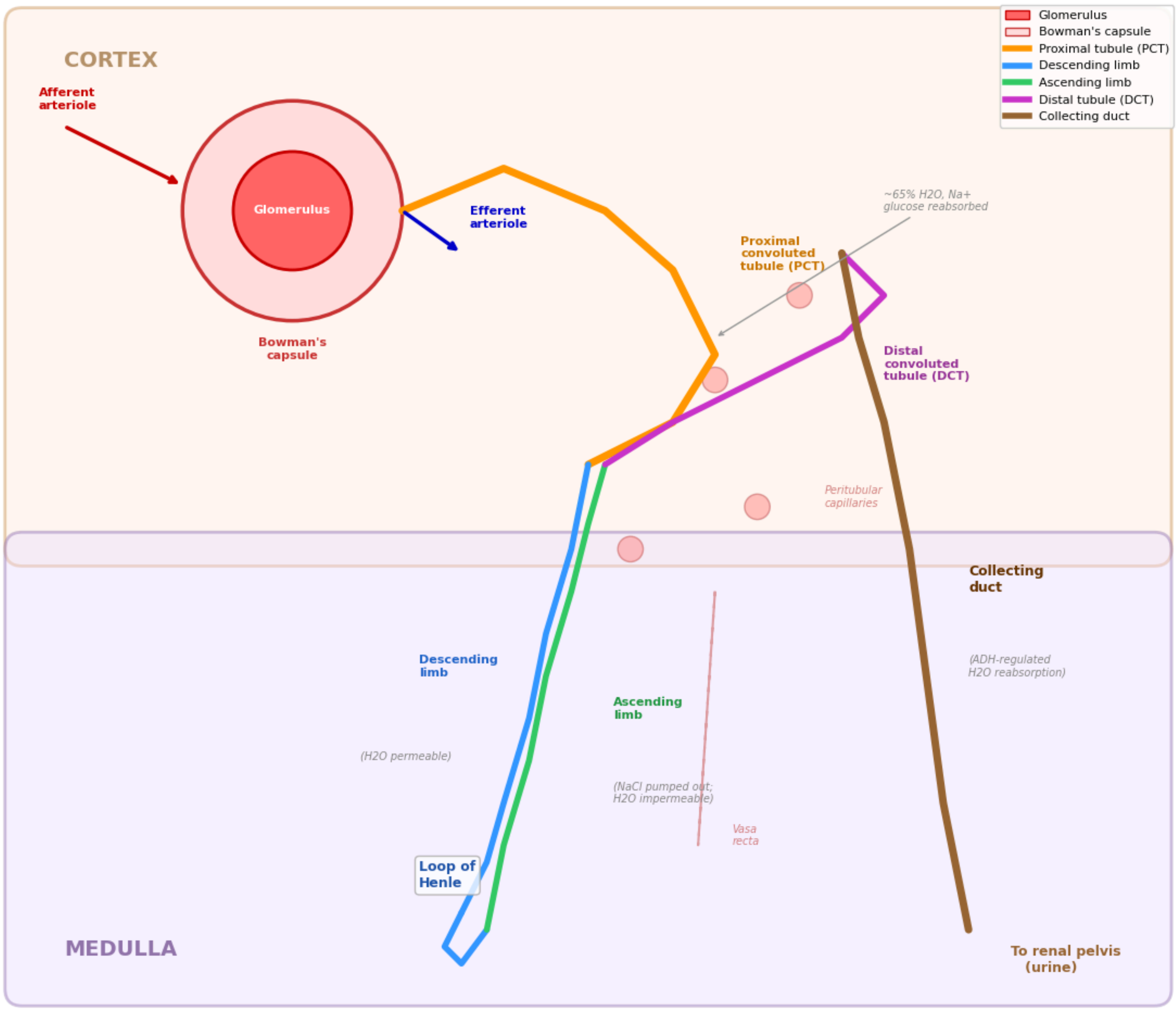
The kidney has a unique **double capillary bed** arrangement:

Renal artery $\rightarrow$ segmental arteries $\rightarrow$ interlobar $\rightarrow$ arcuate $\rightarrow$ **afferent arteriole** $\rightarrow$ **glomerular capillaries** (1st bed: filtration) $\rightarrow$ **efferent arteriole** $\rightarrow$ **peritubular capillaries / vasa recta** (2nd bed: reabsorption) $\rightarrow$ renal vein

This arrangement allows independent regulation of:

- **Glomerular filtration pressure** (by adjusting afferent/efferent arteriolar tone)
- **Peritubular reabsorption** (by adjusting Starling forces in peritubular capillaries)

Figure 2. Anatomy of the Nephron -- The Functional Unit of the Kidney



KEY FEATURES OF THE NEPHRON:

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Each kidney has ~1 million nephrons

Glomerulus: high-pressure filtration (GFR ~120 mL/min)

PCT: bulk reabsorption (~65% of filtered load)

Loop of Henle: counter-current concentrating mechanism

DCT + Collecting duct: fine-tuning under hormonal control

The nephron spans BOTH cortex and medulla

Figure 3a. Starling Forces in Glomerular Filtration

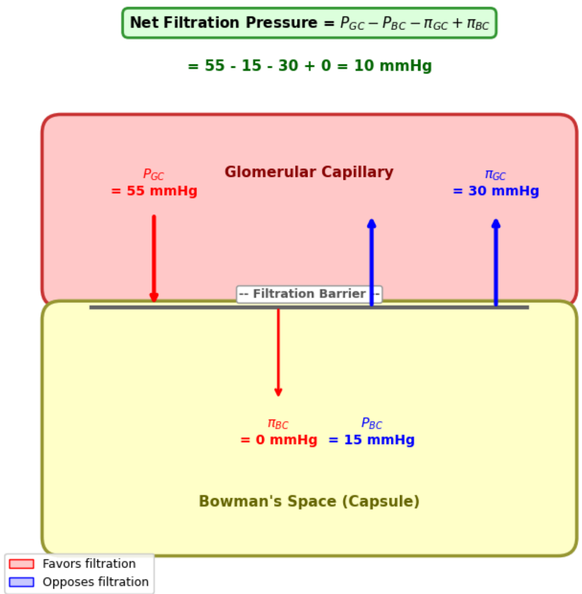
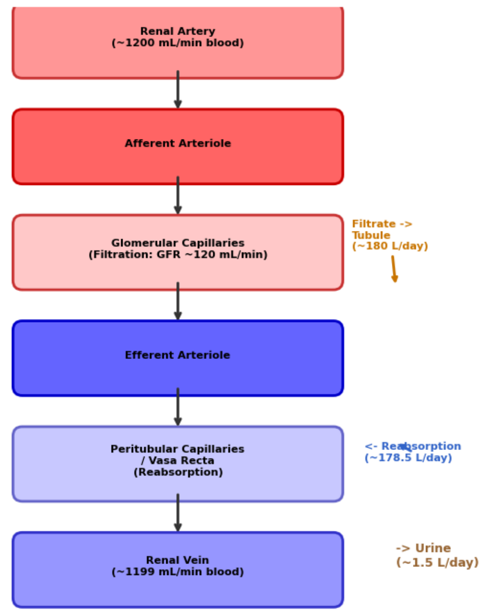


Figure 3b. Renal Blood Flow Pathway



Check Your Understanding: Kidney Anatomy & Physiology

Q1. The kidneys receive ~20--25% of cardiac output, yet they constitute less than 0.5% of body weight. Why do the kidneys need such a disproportionately large blood supply?

▶ Click to reveal answer

Q2. What would happen to GFR if the *afferent* arteriole constricts? What about if the *efferent* arteriole constricts?

▶ Click to reveal answer

Q3. A patient's urine test shows significant amounts of albumin (a large protein, MW ~66 kDa). What does this suggest about the glomerular filtration barrier?

▶ Click to reveal answer

4. Scientific and Clinical Implications

Understanding kidney filtration and GFR has profound implications across medicine and biomedical science:

Chronic Kidney Disease (CKD)

- CKD affects **~10--15% of the global population** (~850 million people) and is a leading cause of death worldwide
- CKD is staged entirely based on **GFR** and **albuminuria**:

Stage	GFR (mL/min/1.73 m^2)	Description
G1	$\geq$ 90	Normal or high (with other evidence of kidney damage)
G2	60--89	Mildly decreased
G3a	45--59	Mildly to moderately decreased
G3b	30--44	Moderately to severely decreased
G4	15--29	Severely decreased
G5	< 15	Kidney failure (may need dialysis or transplant)

Drug Dosing

- The kidney excretes many drugs and their metabolites. When GFR declines, drugs accumulate, causing **toxicity**
- **Dose adjustments** based on eGFR are mandatory for medications such as:
  - Antibiotics (gentamicin, vancomycin)
  - Anticoagulants (enoxaparin, dabigatran)
  - Diabetes medications (metformin --- contraindicated if eGFR < 30)
  - Contrast agents (risk of contrast-induced nephropathy)

Dialysis & Transplantation

- When GFR drops below ~10--15 mL/min (**end-stage renal disease**, ESRD), the kidneys can no longer maintain homeostasis
- **Hemodialysis** uses an external semipermeable membrane and a countercurrent dialysate to mimic glomerular filtration --- sessions of 3--4 hours, 3 times per week
- **Peritoneal dialysis** uses the peritoneal membrane as the filtration surface
- **Kidney transplantation** remains the gold standard for ESRD but is limited by organ availability

Imaging-Based GFR Measurement

- **Nuclear medicine**: ^{99m}Tc-DTPA or ^{51}Cr-EDTA clearance provides accurate GFR measurement
- **Dynamic contrast-enhanced MRI (DCE-MRI)**: Non-invasive measurement of single-kidney GFR using Gd-based contrast agents and pharmacokinetic modeling --- an active area of research that directly relates to computational modeling
- **Renal scintigraphy**: ^{99m}Tc-MAG3 renogram for split renal function assessment

The Kidney in Systemic Disease

- **Diabetes mellitus**: Leading cause of CKD; hyperglycemia damages the glomerular filtration barrier leading to diabetic nephropathy
- **Hypertension**: Both a cause and consequence of CKD; the RAAS is a key therapeutic target (ACE inhibitors, ARBs)

- **Heart failure**: Cardiorenal syndrome --- kidney and heart failure exacerbate each other through hemodynamic and neurohormonal mechanisms

Figure 4a. CKD Stages by GFR

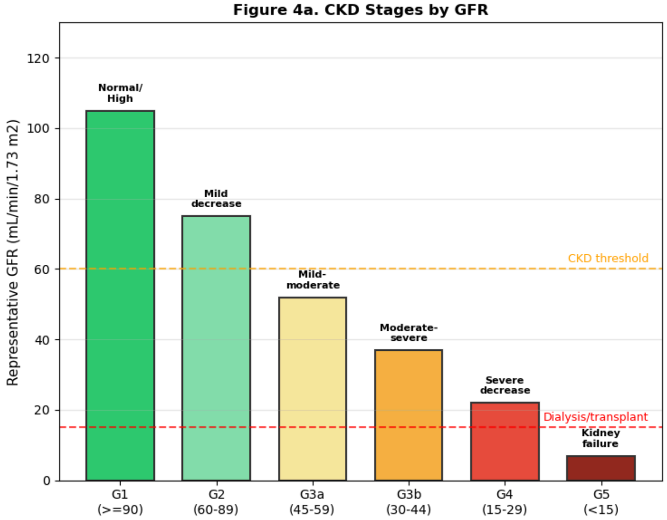
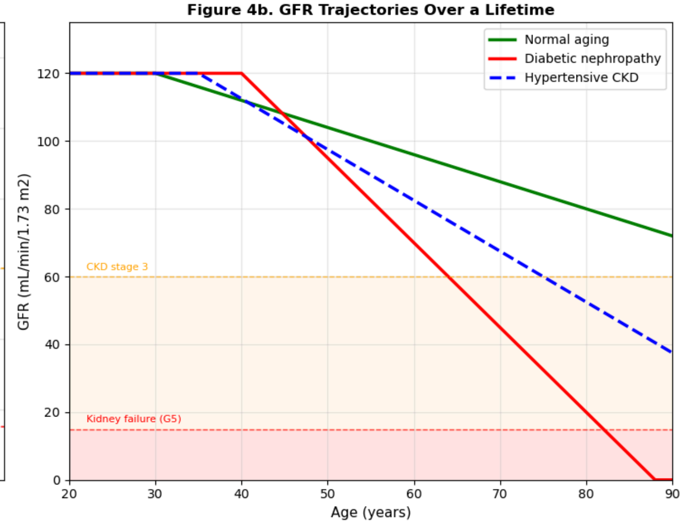


Figure 4b. GFR Trajectories Over a Lifetime

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CLINICAL KEY POINTS:
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  Normal GFR ~ 120 mL/min/1.73 m2 (young adults)
  GFR naturally declines ~0.8 mL/min/year after age 30
  CKD is diagnosed when GFR < 60 mL/min/1.73 m2
  Dialysis is typically needed when GFR < 10-15
  Diabetes and hypertension accelerate GFR decline
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5. Key Definitions and Concepts

Glomerular Filtration Rate (GFR)

GFR is the volume of plasma filtered from the glomerular capillaries into Bowman's capsule per unit time. It is the **gold standard measure of kidney function**.

$$\boxed{\mathrm{GFR} = K_f \times \underbrace{[(P_{GC} - P_{BC}) - (\pi_{GC} - \pi_{BC})]}_{\text{Net filtration pressure (NFP)}}}$$

where:

- K_f = ultrafiltration coefficient (hydraulic permeability $\times$ surface area) [mL/min/mmHg]
- P_{GC} = glomerular capillary hydrostatic pressure (~55 mmHg)
- P_{BC} = Bowman's capsule hydrostatic pressure (~15 mmHg)
- π_{GC} = glomerular capillary oncotic (colloid osmotic) pressure (~30 mmHg)
- π_{BC} = Bowman's capsule oncotic pressure (~0 mmHg, since ultrafiltrate is protein-free)

Normal value: ~120 mL/min/1.73 m^2 (= ~180 L/day)

Renal Plasma Flow (RPF)

The volume of plasma flowing through the kidneys per unit time. Measured using **PAH clearance** (since PAH is nearly completely extracted in a single pass):

$$\text{RPF} \approx \frac{U_{PAH} \times V}{P_{PAH}} \approx 600 \text{ mL/min}$$

Filtration Fraction (FF)

The fraction of renal plasma flow that is filtered at the glomerulus:

$$\boxed{FF = \frac{\text{GFR}}{\text{RPF}} \approx \frac{120}{600} = 0.20 \quad (20\%)}$$

Clearance

The **clearance** of a substance X is the volume of plasma *completely cleared* of X by the kidneys per unit time:

$$\boxed{C_X = \frac{U_X \times \dot{V}}{P_X}}$$

where U_X is the urine concentration, $\dot{V}$ is the urine flow rate, and P_X is the plasma concentration.

Key principle: If substance X is *freely filtered, not reabsorbed, and not secreted*, then:

$$C_X = \text{GFR}$$

This is the basis for measuring GFR using **inulin clearance** (the gold standard) or estimating it from **creatinine clearance**.

Substance handling by the nephron

Substance	Filtered?	Reabsorbed?	Secreted?	Clearance vs. GFR
Inulin	Yes (freely)	No	No	$C_{inulin} = \text{GFR}$
Glucose	Yes	Yes (100%, normally)	No	$C_{glucose} = 0$ (normally)
Creatinine	Yes	No	Slightly	$C_{creatinine} \approx \text{GFR}$ (overestimates slightly)
PAH	Yes	No	Yes (completely)	$C_{PAH} = \text{RPF}$
Urea	Yes	Partially reabsorbed	No	$C_{urea} < \text{GFR}$
Albumin	No (too large)	---	---	$C_{albumin} \approx 0$ (normally)

Check Your Understanding: Key Concepts

Q4. A substance has a clearance *greater* than GFR (i.e., $C_X > C_{inulin}$). What does this tell you about how the nephron handles this substance?

▶ Click to reveal answer

Q5. A patient's serum creatinine level doubles from 1.0 to 2.0 mg/dL. Approximately how much has their GFR changed?

▶ Click to reveal answer

Q6. Why can't glucose clearance be used to measure GFR?

▶ Click to reveal answer

6. Mathematical Framework: Mass Balance and Filtration Modeling

Mass balance for the kidney

The kidney can be modeled as a "black box" with inputs and outputs. For any substance X, conservation of mass at **steady state** requires:

$$\boxed{\underbrace{P_{X,a} \times \mathrm{RPF}_a}_{\text{Arterial input}} = \underbrace{P_{X,v} \times \mathrm{RPF}_v}_{\text{Venous output}} + \underbrace{U_X \times \dot{V}}_{\text{Urine output}}}$$

where:

- $P_{X,a}$, $P_{X,v}$ = arterial and venous plasma concentrations of X [mmol/mL]
- RPF_a, RPF_v = arterial and venous renal plasma flow [mL/min]
- U_X = urine concentration of X [mmol/mL]
- $\dot{V}$ = urine flow rate [mL/min]

Note: $\mathrm{RPF}_v = \mathrm{RPF}_a - \dot{V}$ (the venous plasma flow is reduced by the volume lost as urine).

Clearance derivation from mass balance

The **renal extraction** of X is:

$$E_X = \frac{P_{X,a} - P_{X,v}}{P_{X,a}} = \frac{U_X \times \dot{V}}{P_{X,a} \times \mathrm{RPF}_a}$$

The **clearance** is then:

$$C_X = E_X \times \mathrm{RPF}_a = \frac{U_X \times \dot{V}}{P_{X,a}}$$

The Starling equation for glomerular filtration

The rate of fluid filtration across the glomerular capillary wall is governed by the **Starling equation**:

$$J_v = K_f \left[(P_{GC} - P_{BC}) - \sigma(\pi_{GC} - \pi_{BC}) \right]$$

where:

- J_v = volume filtration rate per unit area [mL/min/cm^2]
- K_f = ultrafiltration coefficient [mL/min/mmHg]
- σ = reflection coefficient (≈ 1 for proteins, ≈ 0 for small solutes)

For the whole glomerulus:

$$\mathrm{GFR} = K_f \times \mathrm{NFP}$$

where NFP = net filtration pressure. With typical values:

$$\mathrm{GFR} = K_f \times [(55 - 15) - (30 - 0)] = K_f \times 10 \text{ mmHg}$$

For GFR = 120 mL/min, this gives $K_f \approx 12$ mL/min/mmHg --- about 100 times higher than in most other capillary beds, reflecting the kidney's specialized filtration barrier.

Dynamic model: GFR along the glomerular capillary

In reality, the oncotic pressure π_{GC} *increases* along the length of the capillary as water is filtered out and plasma proteins become more concentrated:

$$\frac{d\pi_{GC}}{dx} = \frac{\pi_{GC}}{1 - \mathrm{FF}(x)} \times \frac{d\mathrm{FF}}{dx}$$

$$\frac{d\mathrm{FF}}{dx} = \frac{K_f'}{RPF} [P_{GC} - P_{BC} - \pi_{GC}(x)]$$

where K_f' is the ultrafiltration coefficient per unit length and x is the position along the capillary. This means that **filtration pressure decreases along the capillary** and may reach zero before the end (a state called **filtration equilibrium**).

Filtered load, excretion, and net handling

For a given substance X:

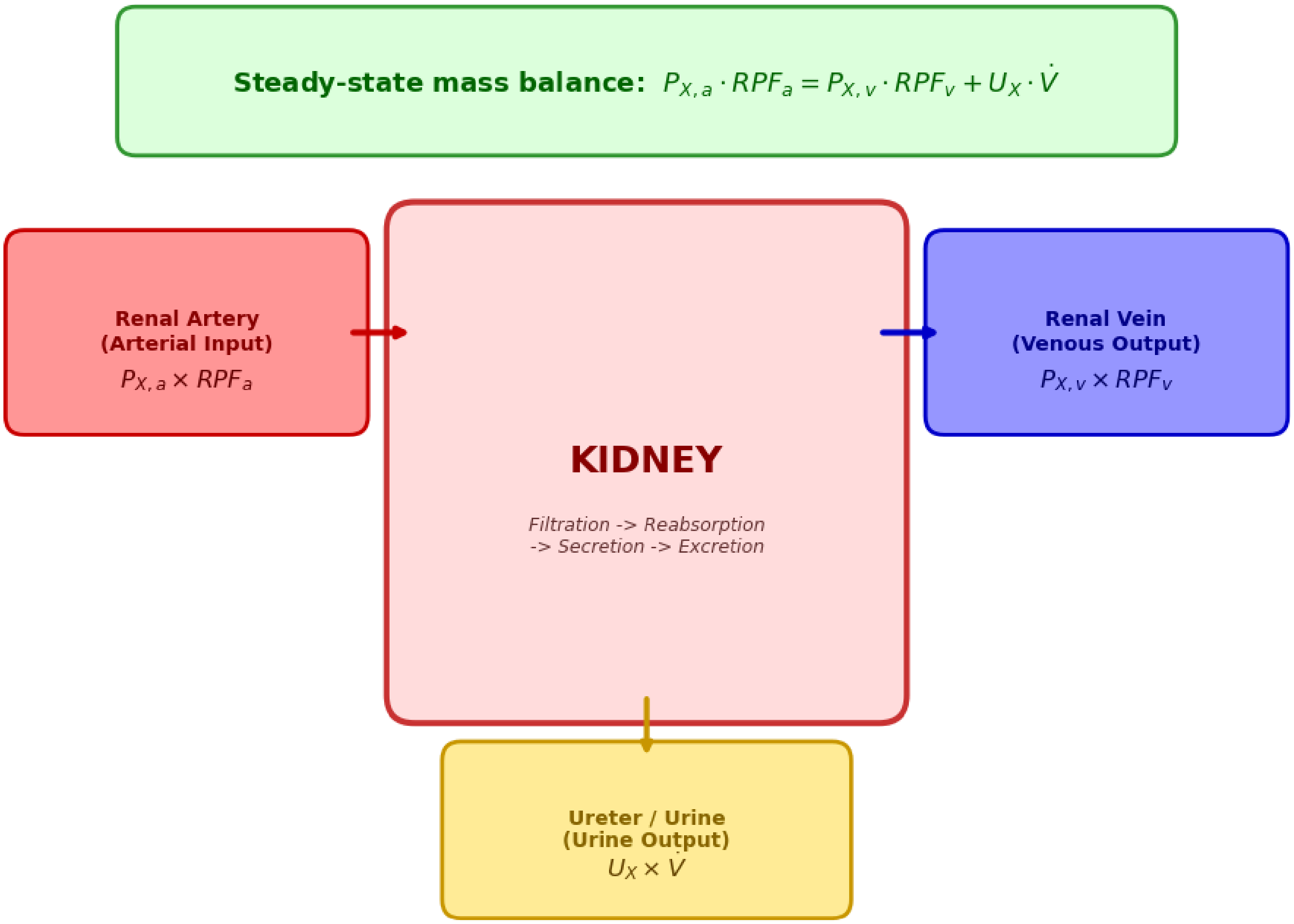
$$\text{Filtered load} = F_X = \mathrm{GFR} \times P_X$$

$$\text{Excretion rate} = E_X = U_X \times \dot{V}$$

$$\text{Net reabsorption} = F_X - E_X \quad \text{(if positive)}$$

$$\text{Net secretion} = E_X - F_X \quad \text{(if positive)}$$

Figure 5. Kidney Mass Balance Schematic for Substance X



Check Your Understanding: The Mathematics

Q7. A patient has the following measurements: plasma inulin concentration P_{in} = 0.5 mg/mL, urine inulin concentration U_{in} = 60 mg/mL, and urine flow rate $\dot{V}$ = 2 mL/min. Calculate the GFR.

▶ Click to reveal answer

Q8. Using the Starling equation, what would happen to GFR if a patient develops severe hypoalbuminemia (very low plasma albumin, e.g., from nephrotic syndrome)?

▶ Click to reveal answer

Q9. (DIY --- Calculation) A patient produces 1440 mL of urine per day with a creatinine concentration of 100 mg/dL. Their serum creatinine is 1.0 mg/dL. Calculate: (a) urine flow rate in mL/min, (b) creatinine clearance, (c) how this compares to normal GFR.

▶ Click for hints
▶ Click to reveal answer

7. Python Implementation: Kidney Filtration Model

We now implement a computational model of kidney filtration based on the mass balance and Starling equation framework developed above. The model includes:

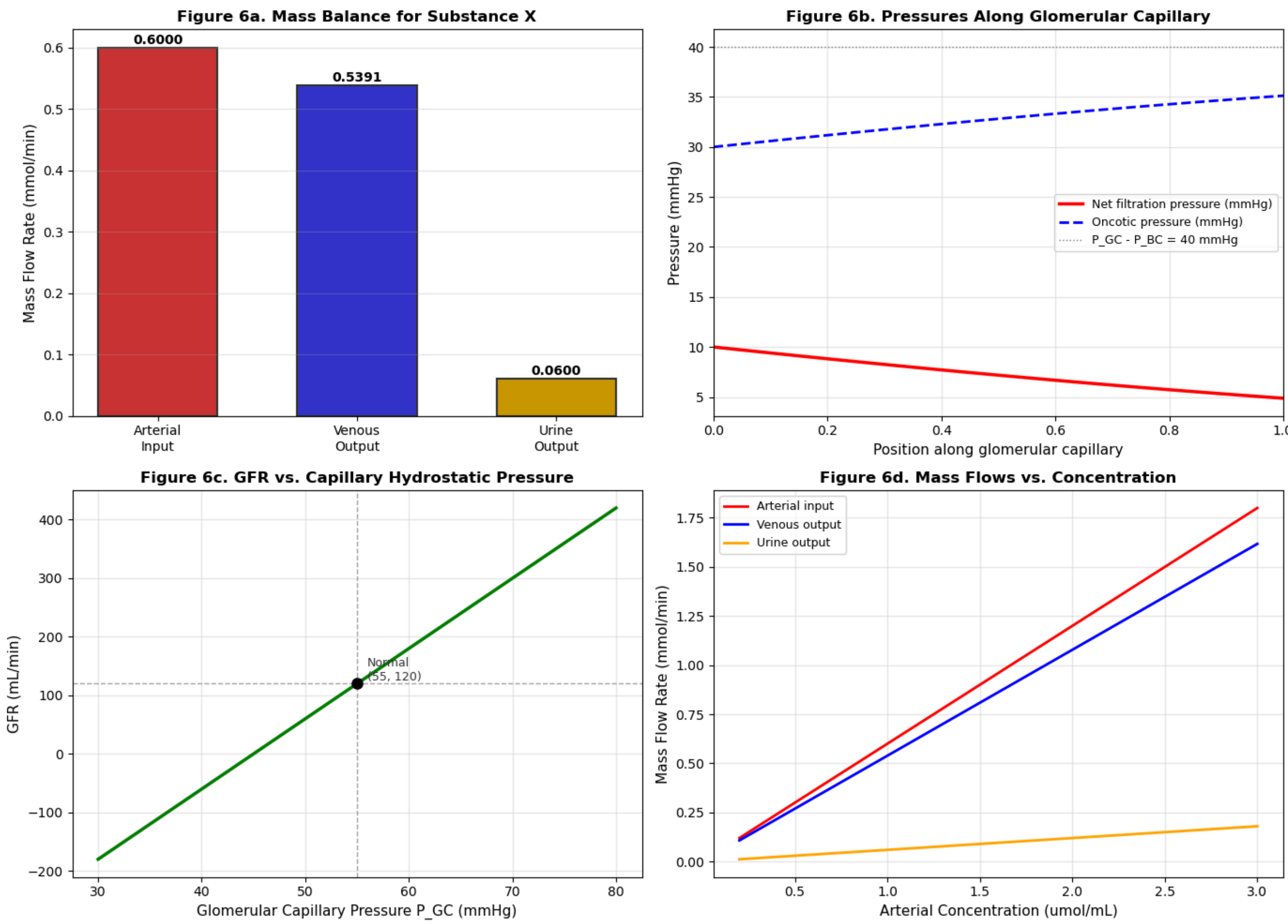
1. **Static mass balance** --- calculating steady-state substance handling
2. **Starling filtration** --- computing GFR from hemodynamic pressures
3. **Dynamic model** --- simulating how oncotic pressure changes along the glomerular capillary
4. **Clearance calculator** --- determining GFR from inulin/creatinine clearance data

```
============================================================
  KIDNEY FILTRATION MODEL -- BASELINE RESULTS
============================================================

  Hemodynamics:
    Renal plasma flow (arterial):  600 mL/min
    Renal plasma flow (venous):    599 mL/min
    GFR:                           120 mL/min
    Filtration fraction:           20.0%

  Starling forces:
    P_GC  = 55 mmHg  (capillary hydrostatic)
    P_BC  = 15 mmHg  (Bowman's hydrostatic)
    pi_GC = 30 mmHg  (capillary oncotic)
    pi_BC = 0 mmHg   (Bowman's oncotic)
    NFP   = 10 mmHg
    K_f   = 12.0 mL/min/mmHg

  Mass balance (for substance X):
    Arterial input:    0.6000 mmol/min
    Venous output:     0.5391 mmol/min
    Urine output:      0.0600 mmol/min
    Balance (~ 0):     0.000900 mmol/min
    Extraction ratio:  10.1%
    Filtered load:     0.1200 mmol/min
    Clearance of X:    60.0 mL/min
============================================================
```




```
KEY OBSERVATIONS:
-------------------------------------------------------
  Panel (a): Mass balance is conserved (input = outputs)
  Panel (b): Oncotic pressure rises along the capillary
    as proteins concentrate, reducing NFP towards zero
  Panel (c): GFR increases linearly with P_GC (Starling)
  Panel (d): Mass flows scale linearly with concentration
```

Interpreting the Simulation Results (Figure 6)

Panel (a) --- Mass Balance: The bar chart confirms that the total arterial input of substance X equals the sum of venous and urine outputs. This is the fundamental mass conservation principle.

Panel (b) --- Filtration Along the Capillary: As blood flows through the glomerular capillary, water is filtered out and plasma proteins become more concentrated, causing oncotic pressure (π_{GC}) to rise. This progressively reduces the net filtration pressure. In some conditions, filtration may cease before the end of the capillary (filtration equilibrium).

Panel (c) --- GFR vs. Pressure: The linear relationship between P_{GC} and GFR (from the Starling equation) shows why blood pressure regulation is critical for kidney function. A drop in P_{GC} (e.g., from hemorrhage or heart failure) directly reduces GFR.

Panel (d) --- Concentration Dependence: For a substance handled passively (no transport maximum), mass flows scale linearly with arterial concentration. In reality, substances like glucose have a **transport maximum (T_m)** --- beyond which they "spill" into the urine.

Interactive Glomerular Filtration Simulator

Use the sliders below to explore how **Starling forces** and the **ultrafiltration coefficient** determine GFR. The left panel shows the instantaneous Starling balance; the right panel shows how oncotic pressure rises and net filtration pressure falls along the glomerular capillary as plasma is concentrated.

The simulation is governed by the **Starling equation**:

$$\text{GFR} = K_f \times \underbrace{[(P_{GC} - P_{BC}) - (\pi_{GC} - \pi_{BC})]}_{\text{Net Filtration Pressure (NFP)}}$$

Adjustable parameters

Parameter	Symbol	Description	Normal	Range	Unit
Glomerular capillary hydrostatic pressure	P_{GC}	Blood pressure inside the glomerular capillary; the main driving force for filtration	55	30–80	mmHg
Bowman's capsule hydrostatic pressure	P_{BC}	Fluid pressure in Bowman's space; **opposes** filtration (rises in obstruction)	15	0–30	mmHg
Glomerular capillary oncotic pressure	π_{GC}	Osmotic pressure from plasma proteins; **opposes** filtration (falls in nephrotic syndrome)	30	10–50	mmHg
Bowman's capsule oncotic pressure	π_{BC}	Oncotic pressure of the filtrate; normally ≈ 0 because the filtration barrier excludes proteins	0	0–10	mmHg
Ultrafiltration coefficient	K_f	Product of capillary hydraulic conductivity and surface area; reduced in glomerulonephritis	12	2–25	mL/min/mmHg
Renal plasma flow	RPF	Plasma flow entering the glomerulus; determines how quickly oncotic pressure rises along the capillary	600	200–1000	mL/min

Try the **preset scenarios** to see how common clinical conditions alter filtration dynamics.

```
HTML(value='<b>Preset clinical scenarios:</b>')
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='auto'), style=ButtonStyl…
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h='420px'), max=80.0, min…
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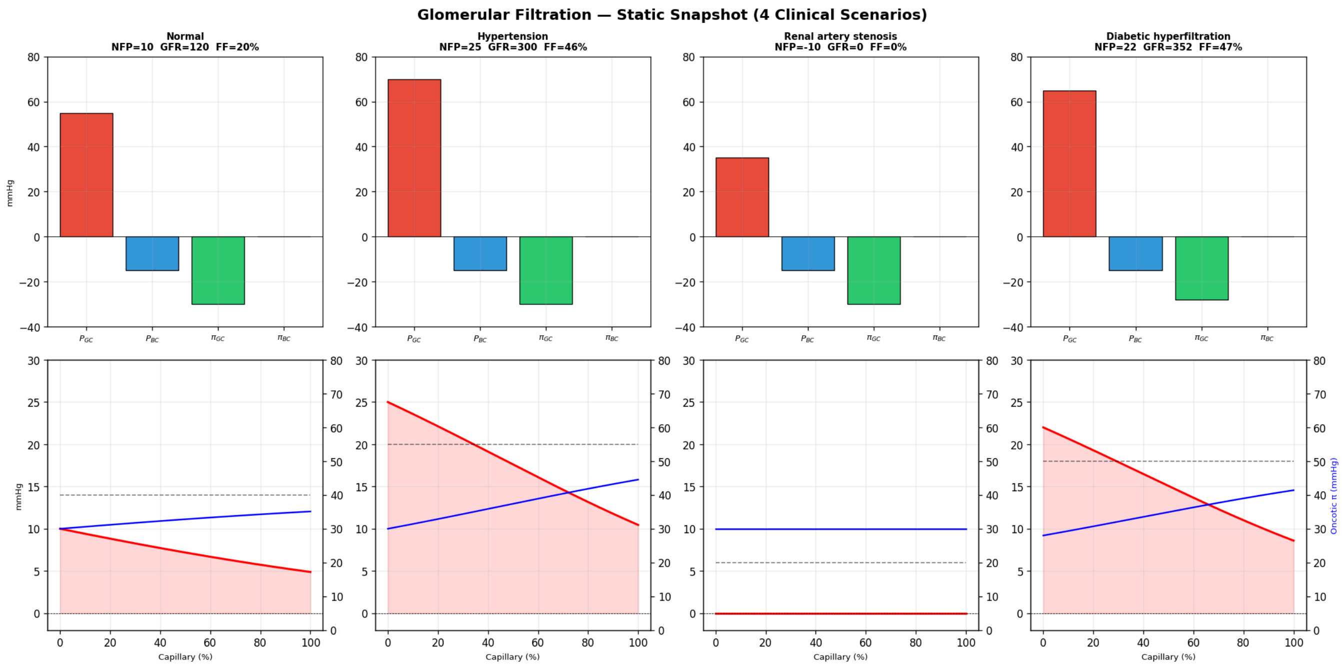


Check Your Understanding: Simulation

Q10. In Figure 6b, what would happen to the oncotic pressure curve if the patient had **hypoalbuminemia** (low plasma albumin)?

▶ Click to reveal answer

Q11. (DIY --- Coding Challenge) Modify the `KidneyParams` to simulate **renal artery stenosis** (narrowing of the renal artery). Reduce `RPF_a` to 300 mL/min and `P_GC` to 35 mmHg. How does this affect GFR, filtration fraction, and mass balance?

▶ Click for starter code

▶ Click to reveal what you should see

8. Clinical GFR Estimation: From Bench to Bedside

Why we need estimated GFR (eGFR)

Measuring true GFR using inulin clearance is the gold standard but requires:

- Continuous IV infusion of inulin
- Timed urine collections (often over 24 hours)
- Multiple blood and urine samples
- It is expensive, time-consuming, and impractical for routine clinical use

In clinical practice, GFR is therefore **estimated** from endogenous biomarkers --- substances naturally present in the blood whose levels reflect kidney function.

Creatinine: the workhorse biomarker

Creatinine is a waste product from muscle metabolism (breakdown of creatine phosphate):

- Produced at a roughly constant rate (~1--2 g/day, proportional to muscle mass)
- Freely filtered at the glomerulus
- Not reabsorbed, but **slightly secreted** (~10--20% overestimation of GFR)
- Steady-state plasma level depends on the balance between production and clearance

The CKD-EPI Equation (2021)

The current clinical standard for eGFR is the **CKD-EPI (Chronic Kidney Disease Epidemiology Collaboration)** equation, updated in 2021 to remove the race coefficient:

$$\mathrm{eGFR} = 142 \times \min(S_{cr}/\kappa, 1)^{\alpha} \times \max(S_{cr}/\kappa, 1)^{-1.200} \times 0.9938^{\mathrm{Age}} \times F_{sex}$$

where:

- S_{cr} = serum creatinine [mg/dL]
- κ = 0.7 (female) or 0.9 (male)
- α = -0.241 (female) or -0.302 (male)
- F_{sex} = 1.012 (female) or 1.0 (male)

Cystatin C: an alternative biomarker

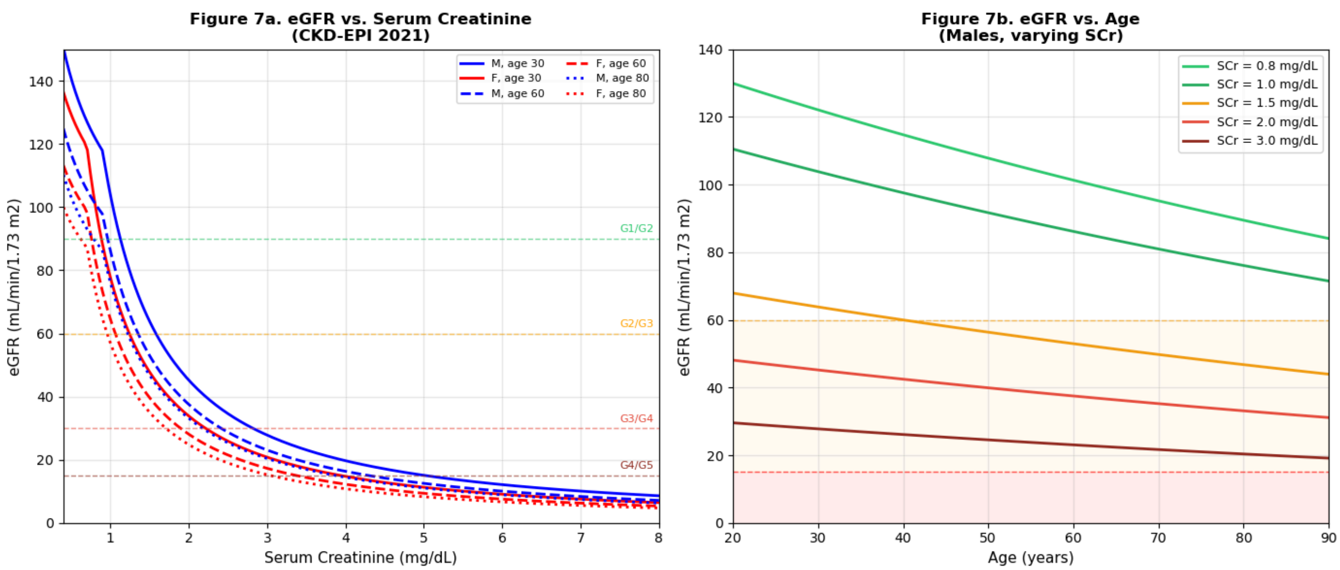
Cystatin C is a small protein produced at a constant rate by all nucleated cells:

- Less influenced by muscle mass, diet, or race than creatinine
- Freely filtered, completely reabsorbed and catabolized by tubular cells (not appearing in urine)
- Useful when creatinine-based estimates may be unreliable (e.g., in patients with abnormal muscle mass)

The CKD-EPI cystatin C equation provides an alternative estimate, and combined creatinine-cystatin C equations offer the best accuracy.

```
=================================================================
  eGFR CALCULATOR -- CKD-EPI 2021 (Race-Free)
=================================================================

  Description                      SCr   Age   Sex    eGFR  Stage
  ---------------------------------------------------------------------------------
--
  Healthy 25y female               0.8    25    F    104.8  G1 (Normal or high)
  Healthy 25y male                 1.0    25    M    107.1  G1 (Normal or high)
  Healthy 60y female               0.9    60    F     73.2  G2 (Mildly decreased)
  Healthy 60y male                 1.1    60    M     76.9  G2 (Mildly decreased)
  Mild CKD, 55y male               1.5    55    M     54.6  G3a (Mild-moderate decreas
e)
  Moderate CKD, 70y female         2.0    70    F     26.4  G4 (Severely decreased)
  Severe CKD, 65y male             4.0    65    M     15.8  G4 (Severely decreased)
  Kidney failure, 50y male         8.0    50    M      7.6  G5 (Kidney failure)
=================================================================
```




```
KEY INSIGHTS:
-------------------------------------------------------
  The eGFR-creatinine relationship is HYPERBOLIC
  -> A doubling of SCr ~ halving of GFR
  Early CKD: large GFR drop causes small SCr rise
    (SCr is a POOR early screening test!)
  Late CKD: small GFR drop causes large SCr rise
  eGFR decreases with age (aging kidney)
  Sex differences reflect differences in muscle mass
```

Check Your Understanding: Clinical GFR Estimation

Q12. A 45-year-old male bodybuilder has a serum creatinine of 1.5 mg/dL. His eGFR (CKD-EPI) calculates to ~55 mL/min, suggesting CKD stage G3a. Is this likely accurate? What might you do next?

▶ Click to reveal answer

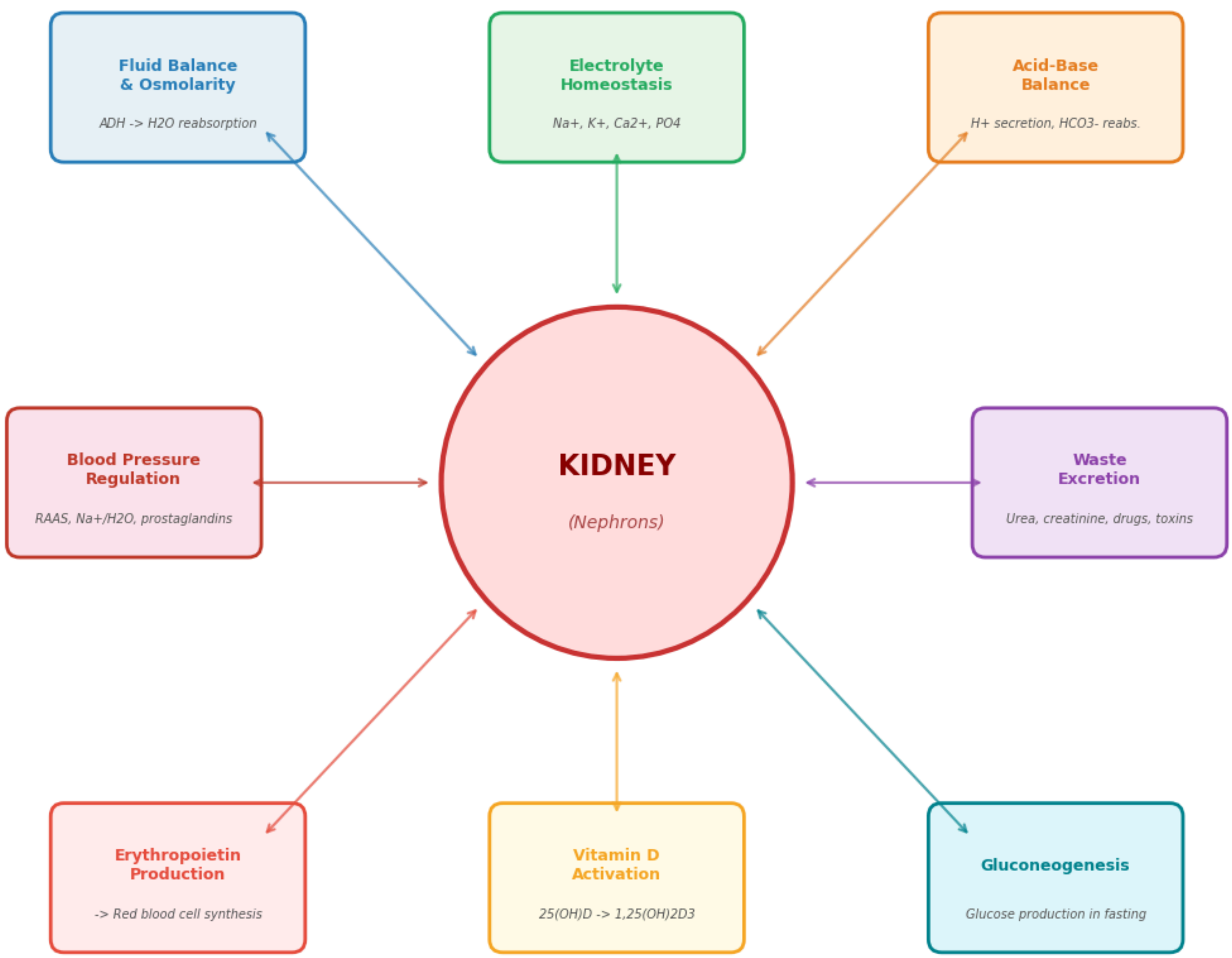
Q13. Why was the race coefficient removed from the CKD-EPI equation in 2021?

▶ Click to reveal answer

Q14. (DIY) Using the `egfr_ckd_epi_2021` function, create a plot showing how eGFR changes with age for a person whose serum creatinine is 1.0 mg/dL. Compare male vs. female trajectories. At what age does each cross into CKD stage G2 territory (eGFR < 90)?

▶ Click for starter code

Figure 8. The Kidney as Master Regulator: Homeostatic Functions



```
THE KIDNEY: NOT JUST A FILTER!
-------------------------------------------------------
  Regulates body fluid volume and osmolarity (ADH)
  Controls electrolyte levels (Na+, K+, Ca2+, PO4)
  Maintains acid-base balance (H+/HCO3-)
  Regulates blood pressure (RAAS system)
  Produces erythropoietin (-> red blood cell production)
  Activates vitamin D (calcium metabolism)
  Performs gluconeogenesis during fasting
  Excretes metabolic waste and drugs
```

9. DIY Exercises: Explore Further

Exercise 1: Diabetes and Hyperfiltration

In early diabetic nephropathy, the GFR is paradoxically **elevated** (hyperfiltration, GFR > 140 mL/min) before it begins to decline. Modify the model to simulate this:

▶ Click for starter code

▶ Click to reveal what you should observe

Exercise 2: Sensitivity Analysis --- What Parameters Most Affect GFR?

▶ Click for starter code

▶ Click to reveal interpretation

Exercise 3: Drug Dosing Calculator

▶ Click for starter code

10. Going Deeper

Extensions and advanced topics

The kidney filtration model presented here can be extended in many directions:

- **Tubuloglomerular feedback (TGF)**: A negative feedback loop where NaCl concentration at the macula densa regulates afferent arteriolar tone. This can be modeled as a delay differential equation
- **RAAS modeling**: The renin-angiotensin-aldosterone system involves multiple compartments and feedback loops --- amenable to systems pharmacology modeling
- **Compartmental pharmacokinetic models**: Drug distribution and elimination through the kidney can be modeled using multi-compartment ODE systems
- **Dynamic contrast-enhanced MRI (DCE-MRI)**: Model-based analysis of Gd-contrast enhancement curves to estimate single-kidney GFR non-invasively. Uses tracer kinetic models

(Tofts, two-compartment exchange models)

- **Whole-kidney simulation**: Combining models of individual nephrons (with heterogeneous properties) to simulate whole-kidney function --- a computational challenge similar to multi-scale cardiac modeling
- **Machine learning for eGFR**: Neural networks trained on large clinical datasets may improve GFR prediction beyond traditional equations

Key references

1. **Smith, H.W.** (1951). *The Kidney: Structure and Function in Health and Disease*. Oxford University Press. [The classic textbook by the father of renal physiology]
2. **Levey, A.S. et al.** (2009). A new equation to estimate glomerular filtration rate. *Ann. Intern. Med.* 150:604--612. [The CKD-EPI equation]
3. **Inker, L.A. et al.** (2021). New creatinine- and cystatin C-based equations to estimate GFR without race. *N. Engl. J. Med.* 385:1737--1749. [The race-free CKD-EPI 2021 update]
4. **Deen, W.M., Robertson, C.R. & Brenner, B.M.** (1972). A model of glomerular ultrafiltration in the rat. *Am. J. Physiol.* 223:1178--1183. [Mathematical model of glomerular filtration]
5. **KDIGO Clinical Practice Guidelines** for CKD: https://kdigo.org/guidelines/ckd-evaluation-and-management/
6. **Boron, W.F. & Boulpaep, E.L.** (2017). *Medical Physiology*, 3rd ed. Elsevier. [Chapters 33--40 on renal physiology]

Useful online resources

- NIDDK --- The Kidneys and How They Work: https://www.niddk.nih.gov/health-information/kidney-disease
- Kidney Disease: Improving Global Outcomes (KDIGO): https://kdigo.org/
- CKD-EPI online calculator: https://www.kidney.org/professionals/kdoqi/gfr_calculator
- PhysioNet --- Renal physiology resources: https://physionet.org/

Connections to other Lab 5 notebooks

- **01-action-potentials**: The kidney's tubuloglomerular feedback uses neural-like signaling; afferent arteriolar tone is modulated by neural input
- **03-cardiovascular-flow**: Renal blood flow regulation is a key component of cardiovascular physiology
- **05-cell-signaling**: The RAAS system involves receptor-mediated cell signaling cascades
- **07-cybernetics-of-wound-healing**: Kidney homeostasis exemplifies cybernetic feedback control

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Notebook summary: This notebook developed a computational model of kidney filtration, covering the anatomical basis (nephron structure, glomerular filtration barrier), the mathematical framework (mass balance, Starling equation, clearance), clinical applications (CKD staging, eGFR estimation, drug dosing), and advanced topics (DCE-MRI, machine learning). The Python implementation allows interactive exploration of how hemodynamic and pathological changes affect kidney function.

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